Problem Set 5: Ordinary Differential Equations

Readings: Newman Ch. 8

Due: Wednesday, October 21st 11:59 PM

Instructions: Work through each problem, following the instructions for the requirements of the code and their outputs. Some starting code is provided for each problem on the course website. Problem sets should be submitted in one batch, with each problem uploaded as it's own python script (`.py` file) on the HuskyCT assignment page with the filename: `ps[PROBLEM SET NUMBER]_[QUESTION NUMBER]_[LASTNAME].py`.

Parameters:

(1) Submitted solutions are restricted to imports of external functions from standard built-in Python libraries (i.e. `math, time, os, sys`, etc.). In addition, imports from `matplotlib.pyplot` and `numpy` are permitted for basic array operations.

(2) All plots should have both axes clearly labeled and include a title. Plots should include either legends or annotations for differentiating plot elements, when applicable.

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1. The Lotka–Volterra equations

The Lotka–Volterra equations are a mathematical model of predator–prey interactions between biological species. Let two variables x and y be proportional to the size of the populations of two species, traditionally called "rabbits" (the prey) and "foxes" (the predators), with x and y as being the population in thousands. Then, $x = 2$ would mean there are 2000 rabbits (as long as x and y do not approach too close to zero, it is okay to have a non-integer number of animals).
In the Lotka–Volterra model the rabbits reproduce at a rate proportional to their population, but are eaten by the foxes at a rate proportional to both their own population and the population of foxes:

$$\frac{dx}{dt} = \alpha x - \beta xy,$$

where α and β are constants. At the same time the foxes reproduce at a rate proportional the rate at which they eat rabbits—because they need food to grow and reproduce—but also die of old age at a rate proportional to their own population:

$$\frac{dy}{dt} = \gamma xy - \delta y,$$

where γ and δ are also constants.
Write a program to solve this system of ODEs for the case $\alpha = 1$, $\beta = \gamma = 0.5$, and $\delta = 2$, starting from the initial condition $x(0) = y(0) = 2$. Your program should:

a. Have a function to solve these ODEs for the time evolution of the populations of rabbits and foxes, $x(t)$, $y(t)$, using the fourth-order Runge-Kutta method.

b. Calculate and print the values of the population levels at $t = 10$ to the nearest integer amount of rabbit or fox.

c. Make a plot of the population level evolution of rabbits and foxes between $t = 0$ to $t = 30$.

d. Save the figure as `LVpop_[LASTNAME].png`

Note that while these equations don't explicitly depend on time, you may still find it useful to define functions to take time as an input in order to iterate in time.

2. VIBRATION IN A ONE-DIMENSIONAL SYSTEM

Let's study the motion of a system of N identical masses (in zero gravity) joined by identical linear springs like this:



As shown in Newman's Example 6.2, the horizontal displacements ξ_i of masses $i = 1 \ldots N$ satisfy equations of motion

$$m\frac{d^2\xi_1}{dt^2} = k(\xi_2 - \xi_1) + F_1,$$
$$m\frac{d^2\xi_i}{dt^2} = k(\xi_{i+1} - \xi_i) + k(\xi_{i-1} - \xi_i) + F_i,$$
$$m\frac{d^2\xi_N}{dt^2} = k(\xi_{N-1} - \xi_N) + F_N.$$

where m is the mass, k is the spring constant, and F_i is the external force on mass i. Example 6.2 shows how these equations could be solved by guessing a form for the solution and using a matrix method. Here we'll solve them more directly.

Write a program to solve for the motion of N masses using the fourth-order Runge-Kutta method, where $m = 1\ \mathrm{kg}$, $k = 6\ \mathrm{kg\ s^{-2}}$ and the driving forces are all zero except for $F_1 = \cos\omega t$ with $\omega = 2\ \mathrm{s^{-1}}$. Recall: you will need to convert each second-order equations of motion a first-order equation. Your program should:

a. Have a function that calculates displacement for a general number of N masses as input using the RK4 method. The function should have the RK4 method operate on a vector of length $2N$.

b. Calculate and plot the solutions for $N = 5$ masses from $t = 0$ to $t = 20$ on the same panel, assuming that all masses start at rest with zero displacement: $\xi_i = 0$ and $d\xi_i/dt = 0$ at $t = 0$.

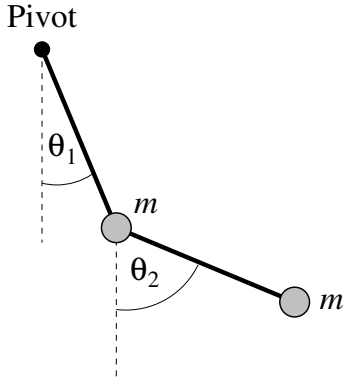
c. Analyze the displacement history to compute and print the time to within a hundredth of a second at which all masses appear to return their original positions (within $\sim 2\ \mathrm{cm}$).

Hint: You may wish to restrict the search for $t > 2.5\ \mathrm{s}$

d. Save the figure as `springs_[LASTNAME].png`.

3. CHAOTIC DOUBLE PENDULUM

A double pendulum consists of a normal pendulum with another pendulum hanging from its end. For simplicity let us ignore friction, and assume that both pendulums have bobs of the same mass m and massless arms of the same length ℓ. Thus the setup looks like this:



The position of the arms at any moment in time is uniquely specified by the two angles θ_1 and θ_2. The potential energy of the system is given by:

$$V = mgh_1 + mgh_2 = -mg\ell(2\cos\theta_1 + \cos\theta_2),$$

where g is the acceleration due to gravity.

The total kinetic energy is

$$T = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = m\ell^2\left[\dot\theta_1^2 + \frac{1}{2}\dot\theta_2^2 + \dot\theta_1\dot\theta_2\cos(\theta_1 - \theta_2)\right],$$

and thus the Lagrangian of the system is

$$\mathcal{L} = T - V = m\ell^2\left[\dot\theta_1^2 + \frac{1}{2}\dot\theta_2^2 + \dot\theta_1\dot\theta_2\cos(\theta_1 - \theta_2)\right] + mg\ell(2\cos\theta_1 + \cos\theta_2).$$

Then the equations of motion for $\ddot\theta_1$ and $\ddot\theta_2$ can be found by applying the Euler–Lagrange equations. The equations of motion are second-order equations, but we can convert them into first-order ones by the usual method, defining two new variables, ω_1 and ω_2, thus:

$$\dot\theta_1 = \omega_1, \qquad \dot\theta_2 = \omega_2.$$

which then yield the following additional first order ODEs for the equations of motion:

$$\dot\omega_1 = -\frac{\omega_1^2\sin(2\theta_1 - 2\theta_2) + 2\omega_2^2\sin(\theta_1 - \theta_2) + (g/\ell)\left[\sin(\theta_1 - 2\theta_2) + 3\sin\theta_1\right]}{3 - \cos(2\theta_1 - 2\theta_2)},$$
$$\dot\omega_2 = \frac{4\omega_1^2\sin(\theta_1 - \theta_2) + \omega_2^2\sin(2\theta_1 - 2\theta_2) + 2(g/\ell)\left[\sin(2\theta_1 - \theta_2) - \sin\theta_2\right]}{3 - \cos(2\theta_1 - 2\theta_2)}.$$

(With a few trigonometric identities being applied along the way). These two equations, along with the equations $\dot\theta_1 = \omega_1$ and $\dot\theta_2 = \omega_2$, give us four first-order equations which between them define the motion of the double pendulum. There is no analytic closed form solution for $\theta_1(t)$ and $\theta_2(t)$, so they must be solved numerically!
Write a program using the fourth-order Runge-Kutta method to solve the equations of motion for the trajectories of $\theta_1(t), \theta_2(t), \omega_1(t), \omega_2(t)$. Take $\ell = 40$ cm and $m = 1$ kg each. Your program should:

a. Have a function to compute the trajectory based on the equations of motion as a function of the initial displacements $\theta_1(0)$ and $\theta_2(0)$ given some timestep h. Assume that $\omega_1(0) = 0, \omega_2(0) = 0$.

b. Have a function that computes the total energy $E = T + V$ of the system in terms of the variables θ_1, θ_2, ω_1, and ω_2.

c. Compute the evolution of θ and ω from $t = 0$ to $t = 10$ seconds for the two pendulums, given an initial $\theta_1 = 15°$, $\theta_2 = 15°$. Choose a timestep size such that the variation in total energy is less than 10^{-6} J over the course of the entire calculation.

d. Make a plot with two panels. One panel should show $\theta(t)$ for the two pendulums and the exact solution for a single pendulum at the same initial condition: $\theta(t) = \theta_0\cos(\sqrt{g/L}\,t)$ for comparison. The second panel should show ΔE, the change in total energy over time relative to the initial energy, during the calculation from $t = 0$ to $t = 10$ seconds.

e. Save the figure as `doublependulumtime_[LASTNAME].png`.

f. Plot the calculated trajectories from part (b) over time in a *phase-space* (ω vs. θ) diagram. Include the phase space solution for the single pendulum for comparison.

g. Save the figure as `doublependulumphase_[LASTNAME].png`.

4. TRAJECTORY WITH AIR RESISTANCE

Many elementary mechanics problems deal with the physics of objects moving or flying through the air, but they almost always ignore friction and air resistance to make the equations solvable. If we're using a computer, however, we don't need solvable equations.

Consider, for instance, a spherical cannonball shot from a cannon standing on level ground. The air resistance on a moving sphere is a force in the opposite direction to the motion with magnitude

$$F = \frac{1}{2}\pi R^2 \rho C v^2,$$

where R is the sphere's radius, ρ is the density of air, v is the velocity, and C is the so-called *coefficient of drag* (a property of the shape of the moving object, in this case a sphere).

Starting from Newton's second law, $F = ma$, we can show that the equations of motion for the position (x, y) of the cannonball are

$$\ddot{x} = -\frac{\pi R^2 \rho C}{2m}\,\dot{x}\sqrt{\dot{x}^2 + \dot{y}^2}, \qquad \ddot{y} = -g - \frac{\pi R^2 \rho C}{2m}\,\dot{y}\sqrt{\dot{x}^2 + \dot{y}^2},$$

where m is the mass of the cannonball, g is the acceleration due to gravity, and $\dot{x}$ and $\ddot{x}$ are the first and second derivatives of x with respect to time.

Take a standard cannonball to be of mass $1\ \mathrm{kg}$ and radius $8\ \mathrm{cm}$, with the coefficient of drag for a sphere $C = 0.47$. Assume the density of air is $\rho = 1.22\ \mathrm{kg\,m^{-3}}$ and the starting position of the cannonball is on the ground at the origin. Write a program to calculate the trajectory of a cannonball under air resistance. The program should:

a. Define a function for the four first order ODEs to be solved for this system.

b. Include a method to solve for the trajectories $(x(t), y(t))$ given an initial velocity vector, using an ODE solver of your choice.

c. Calculate and print the time at which the cannonball would strike the ground (within $1\ \mathrm{cm}$).

d. Plot the trajectory (y vs. x) for a cannonball launched at a speed $v_0 = 100\ \mathrm{m/s}$ at angle of $\theta = 30^\circ$ from the ground for this cannonball vs. one without air resistance.

e. Save this figure as `projectiles_[LASTNAME].png`.

f. Use your code to investigate how the mass of projectile affects the trajectory, distance traveled, etc. for a cannonball experiencing air resistance. How heavy does a cannonball have to be to be unaffected by air resistance? Make a summary plot to demonstrate and explain your findings save it as `airresist_[LASTNAME].png`.